Dual-Resolution Multi-View Action DETR with Hand-Trajectory Cross-Attention and Existence-Quality Calibration

Authors & Affiliation

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- **Challenge Track:** RetailAction Challenge (PRAW Workshop @ WACV 2027) — Top-View Spatio-Temporal Action Localization
- **Final Public Leaderboard Score:** `0.7942`

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1. Executive Summary

We present a query-based multi-view spatio-temporal action localization framework for dual overhead retail cameras (`rank0` and `rank1`). Starting from a shared `MViTv2-S` video backbone and a 2-layer / 3-layer Transformer encoder-decoder with $Q = 5$ learned action queries (`MultiViewActionDETR`), we systematically investigated why standard DETR-style formulations underperform on the class-agnostic 100-threshold RetailAction metric ($\mathrm{TAU_T} \in [0.05, 0.50] \times \mathrm{TAU_S} \in [0.10, 1.00]$ m).

Our final system (**V4**, Public Leaderboard `0.7942`, Local Validation `0.8541`, Local Test `0.8420`) combines five core components:

1. **Objectness/Class Decoupling & Per-Layer Fresh Hungarian Re-Matching (STMixer-style [5]):** Decouples query existence (`obj_head`, supervised with binary cross-entropy across all 5 queries) from action classification (cross-entropy restricted to matched queries), and re-runs Hungarian matching independently at every decoder layer (`w_aux = 0.4`).
2. **Dense Per-Frame Multi-Label Supervision & All-Query Continuous Quality Calibration (PointTAD [2] / VarifocalNet [1]-style):** Auxiliary frame-level multi-label supervision (`dense_head` [2]) sharpens backbone temporal boundaries, while continuous spatio-temporal IoU quality (`quality_head` [1] + geometry-conditioned `iou_head` with 102-D Fourier tuple encodings) calibrates query confidence.
3. **Residual Hand-Trajectory Cross-Attention (MV-TAP-style [6]):** Extracts 32-frame 2D wrist trajectories, velocities, and visibility flags from the provided per-clip top-view pose JSONs (`extract_hand_tracks`), encoding 128 spatiotemporal hand tokens that refine decoder queries via gated residual cross-attention without diluting visual cross-attention weights [6].
4. **Progressive Multi-Resolution Curriculum (`224×224` → `256×256` → Paired `256×256` + `288×288`):** Scales spatial token grids from 7×7 (98 tokens/view) to 8×8 (128 tokens/view) and 9×9 (162 tokens/view) via bicubic 2D positional-embedding interpolation and activation checkpointing, with corrected `AdamW` weight-decay hygiene [3] and ranking-consistent checkpoint selection.
5. **Slot-Aligned 4-Pass TTA & Dual-Resolution Trajectory Ensembling [4]:** Because the 5 learned query slots preserve stable temporal-spatial specialization across fine-tuning and resolutions [4], we average per-slot predictions across 4 test-time augmentations (`s0`, `s1`, `flip`, `flip1`) and 4 complementary model states (`256px` RAW + EMA [3] and `288px` RAW + EMA [3]) without destructive NMS, followed by a single post-ensemble symmetric temporal boundary expansion (`pad_t = 0.015`).

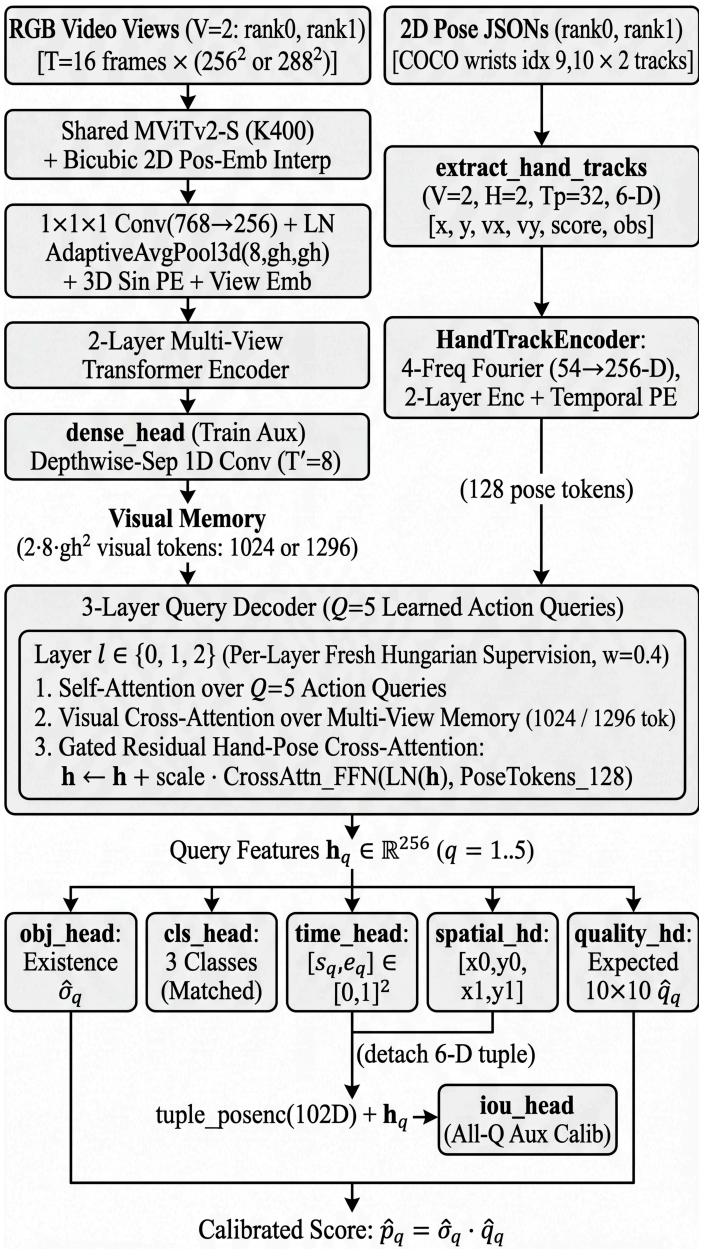
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2. Model Architecture

Given synchronized top-view RGB video streams from two overhead cameras ($V = 2$, `rank0` and `rank1`) and their accompanying 2D human pose detections (`poses_rank0.json`, `poses_rank1.json`), our model `MultiViewActionDETR` predicts $Q = 5$ action tuples per clip:

$$\hat{y}_q = (\hat{s}_q, \hat{e}_q, \hat{x}_{q,0}, \hat{y}_{q,0}, \hat{x}_{q,1}, \hat{y}_{q,1}, \hat{p}_q), \quad q \in \{1, \dots, 5\}$$

where $[\hat{s}_q, \hat{e}_q] \subseteq [0,1]^2$ is the normalized temporal interval, $(\hat{x}_{q,v}, \hat{y}_{q,v}) \in [0,1]^2$ is the normalized 2D interaction point in camera view $v \in \{0,1\}$, and $\hat{p}_q \in (0,1)$ is the calibrated ranking score.



2.1 Shared Video Backbone & Resolution-Adaptive Token Grid

- **Backbone:** `torchvision mvit_v2_s` initialized with Kinetics-400 weights (`MViT_V2_S_Weights.KINETICS400_V1`), taking $T = 16$ uniformly sampled RGB frames per view in chunks of 16.
- **3D Adaptive Spatio-Temporal Pooling:** Backbone tokens are reshaped to (B, C_{768}, T', H', W'), projected via $1 \times 1 \times 1$ convolution + LayerNorm to $d_{\text{model}} = 256$, and pooled via `AdaptiveAvgPool3d((grid_t, grid_hw, grid_hw))` with `grid_t = 8` and resolution-matched spatial grids:
  - `224×224`: `grid_hw = 7` ($8 \times 7 \times 7 = 392$ tokens/view, 784 multi-view tokens total)
  - `256×256`: `grid_hw = 8` ($8 \times 8 \times 8 = 512$ tokens/view, $1,024$ multi-view tokens total)
  - `288×288`: `grid_hw = 9` ($8 \times 9 \times 9 = 648$ tokens/view, $1,296$ multi-view tokens total)
- **Positional & View Embeddings:** 3D sinusoidal spatio-temporal positional embeddings (`sinusoidal_3d`) and learned camera view embeddings (`view_emb`: 2×256) are added to all visual tokens before concatenating both views into a unified multi-view memory sequence. When transitioning between spatial resolutions (`224` → `256` → `288`), the backbone's learned 2D spatial positional embedding `backbone.pos_hw` is bicubically interpolated on-the-fly (`model.py:L26-L31`).

2.2 Multi-View Transformer Encoder-Decoder & Hand-Pose Cross-Attention

- **Visual Encoder:** A 2-layer Pre-LN `TransformerEncoder` ($d = 256$, 8 heads, FFN dim 1024, GELU, dropout 0.1) performs joint self-attention across all spatio-temporal tokens from both camera views.
- **Hand-Trajectory Token Encoder (`HandTrackEncoder` [6]):** Top-view retail actions (shelf reaching, taking/returning items) are tightly coupled to wrist trajectories. From `poses_rank0.json` and `poses_rank1.json`, `extract_hand_tracks` extracts left and right wrist keypoints (COCO indices 9 and 10) for the two highest-confidence person tracks across $T_p = 32$ uniformly sampled time steps per view, forming a $(V = 2, H = 2, T_p = 32, 6)$ tensor of $[x, y, v_x, v_y, \text{score}, \text{obs}]$ with linear interpolation across occluded frames [6]. Each 6-D state is lifted with 4-frequency Fourier positional features (54-D total) and projected to 128 pose tokens of dimension 256.
- **Residual Pose Cross-Attention:** Rather than concatenating pose tokens into the visual memory (which dilutes visual cross-attention softmax weights when warm-starting), each of the 3 Pre-LN `TransformerDecoder` layers applies standard visual cross-attention first, followed by a gated residual cross-attention block over the 128 hand-pose tokens (`scale` initialized to 0.05) [6].

2.3 Prediction & Calibration Heads

From each decoder layer's query representations $h_q \in \mathbb{R}^{256}$ ($q = 1 \dots 5$):

1. **Existence / Objectness Head (`obj_head` [5]):** `Linear(256, 1)` predicting query existence probability $\hat{o}_q = \sigma(z_q^{\text{obj}})$.
2. **Action Class Head (`cls_head` [5]):** `Linear(256, 4)` (3 action classes + auxiliary slot; supervised only on matched queries [5]).
3. **Temporal Head (`time_head`):** 2-layer MLP predicting $(\alpha_q, \beta_q) \in (0,1)^2$, parameterized as $\hat{s}_q = \alpha_q$ and $\hat{e}_q = \alpha_q + (1-\alpha_q)\beta_q$ to guarantee $0 \le \hat{s}_q \le \hat{e}_q \le 1$.
4. **Multi-View Spatial Head (`spatial_head`):** 3-layer MLP predicting normalized 2D coordinates $[\hat{x}_{q,0}, \hat{y}_{q,0}, \hat{x}_{q,1}, \hat{y}_{q,1}] \in (0,1)^4$.
5. **Expected Metric Quality Head (`quality_head` [1]):** 2-layer MLP predicting $\hat{q}_q = \sigma(z_q^{\text{qual}})$, supervised against the exact expected match fraction over the 10×10 challenge evaluation threshold grid.
6. **Geometry-Conditioned All-Query IoU Head (`iou_head` [1]):** Concatenates h_q with a 102-D multi-frequency sinusoidal encoding (`tuple_posenc`) of the detached 6-D coordinate tuple $[\hat{s}_q, \hat{e}_q, \hat{x}_{q,0}, \hat{y}_{q,0}, \hat{x}_{q,1}, \hat{y}_{q,1}]$ and predicts all-query overlap quality across both matched and unmatched queries.
7. **Dense Per-Frame Auxiliary Head (`dense_head` [2], training only):** Pools multi-view encoder memory spatially per temporal bin ($T' = 8$) and applies a depthwise-separable 1D temporal convolution to predict frame-level multi-label action probabilities [2].

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3. Training Data & Preprocessing

- **Dataset:** Official RetailAction Challenge dataset exclusively (**no external video or retail datasets** were used; only standard Kinetics-400 initialization for `mvit_v2_s` from `torchvision`).
  - **Train split:** `2,519` dual-view clips (`annotations/annotations_train.json`)
  - **Validation split:** `1,277` dual-view clips (`annotations/annotations_validation.json`)
  - **Local Test split:** `2,501` dual-view clips (`annotations/annotations_test.json`)
  - **Challenge split:** `2,500` unlabeled dual-view clips (`sample_submission.csv`)
- **Physical Scale Calibration:** Following the challenge specification, per-view pixel-to-meter conversion factors (`meters_per_unit`) are computed from the median limb-length ratios of detected 2D skeletons against anatomical reference lengths (`util.py:L15-L43`), defaulting to 3.5 m when fewer than 3 valid limbs are visible.
- **Training Augmentations:** Random temporal jitter (`±1` frame offset around uniform 16-frame linspace sampling) and independent per-view horizontal flipping ($p = 0.5$), which simultaneously flips the RGB frames (`x -> W - 1 - x`), ground-truth interaction x-coordinates (`x -> 1 - x`), and extracted hand-track x-positions/velocities (`x -> 1 - x`, `vx -> -vx`) while swapping left/right wrist indices.

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4. Training Recipe & Progressive Curriculum

4.1 Multi-Task Loss Formulation

At each of the 3 decoder layers ($l \in \{0,1,2\}$, with layer weights $w_0 = 0.4, w_1 = 0.4, w_2 = 1.0$), we run **fresh Hungarian matching** (`HungarianMatcher` [5]) with cost matrix:

$$\mathcal{C}_{q,m} = w_{\text{cls}} \left(-\hat{P}_{q,c_m}\right) + w_{\text{obj}} \left(-\hat{o}_q\right) + w_s \|\hat{\mathbf{x}}_q - \mathbf{x}_m\|_1 + w_t \left(\|\hat{\mathbf{t}}_q - \mathbf{t}_m\|_1 - \text{GIoU}_t(\hat{\mathbf{t}}_q, \mathbf{t}_m)\right)$$

with $w_{\text{cls}} = 1, w_{\text{obj}} = 2, w_s = 5, w_t = 5$. Given the matched pairs, the per-layer loss is:

$$\mathcal{L}_{\text{layer}} = 2.0\,\mathcal{L}_{\text{obj-BCE}}^{\text{all } Q} + 1.0\,\mathcal{L}_{\text{cls-CE}}^{\text{matched}} + 5.0\,\mathcal{L}_{\text{spatial-}L_1} + 5.0\,\mathcal{L}_{\text{temp-}L_1} + 2.0\,\mathcal{L}_{\text{temp-DIoU}} + 0.2\,\mathcal{L}_{\log\text{-dur}} + 4.0\,\mathcal{L}_{\text{qual-BCE}} + 3.0\,\mathcal{L}_{\text{iou-BCE}}^{\text{all } Q} + 1.0\,\mathcal{L}_{\text{iou-rank}}$$

plus the encoder-level dense per-frame loss $3.0\,\mathcal{L}_{\text{dense-BCE}}$ [2] (active even on zero-GT clips). Crucially, quality targets $q_q^* \in [0,1]$ [1] are computed under `torch.no_grad()` as the exact fraction of the 100 evaluation thresholds (τ_t, τ_s) satisfied by the query:

$$q_q^* = \left(\frac{1}{10}\sum_{\tau_t \in \text{TAU_T}} \mathbb{I}[\text{tIoU}_q > \tau_t]\right) \cdot \left(\frac{1}{10}\sum_{\tau_s \in \text{TAU_S}} \mathbb{I}[\min(d_{q,0}, d_{q,1}) < \tau_s]\right)$$

4.2 Progressive Training Stages

All stages use `AdamW` with weight-decay hygiene [3] (weight decay `1e-4` applied strictly to ≥ 2-D `Linear`/`Conv` weight matrices; `0.0` on all biases, LayerNorms, positional embeddings, and query tokens), `bfloat16` mixed precision, gradient norm clipping at `1.0`, cosine learning rate schedule, and exponential moving average (`ModelEma` [3]).

Stage	Checkpoint	Init From	Resolution (`grid_hw`)	Epochs	Batch × Accum	Base / Backbone / New-Head LR	Key Additions	Val `mAP`	Local Test `mAP`	Publi LI
Host Baseline	—	—	`224` (`7×7`)	—	—	—	Official challenge baseline	—	—	`0.391`
Stage 1 (`v2`)	`run2q`	Kinetics-400	`224` (`7×7`)	12	4×4	`1e-4` / `1e-5` / —	Base `MultiViewActionDETR` + `quality_head` [1]	`0.6171`	`0.5972`	`0.555`
Stage 2 (`idea_07`)	`ab_diou_ema`	`run2q`	`224` (`7×7`)	4	8×4	`5e-5` / `5e-6` / —	Temporal DIoU + `ModelEma(0.999)` + WD-hygiene [3]	`0.7034`	`0.6877`	–

Stage	Checkpoint	Init From	Resolution (grid_hw)	Epochs	Batch × Accum	Base / Backbone / New-Head LR	Key Additions	Val mAP	Local Test mAP	Publi LI
Stage 3 (idea_11)	stmixer_ft	ab_diou_ema	224 (7×7)	3	8×4	5e-5 / 5e-6 / 5e-4	obj_head BCE + matched-only CE + per-layer re-matching (w_aux=0.4) [5]	0.7864	0.7743	–
Stage 4 (v2_comb)	combine_v2a	stmixer_ft (EMA)	224 (7×7)	4	8×4	4e-5 / 4e-6 / 5e-4	dense_head (w=3.0) [2] + wq=4.0 [1] + 4-pass TTA (f4) [4] + noNMS + EMA+RAW [3]	0.8157	0.8082	0.758
Stage 5a (v3a)	combine_v3a	combine_v2a (EMA)	224 (7×7)	4	16×2	3e-5 / 3e-6 / 5e-4	HandTrackEncoder pose cross-attn [6] + iou_head [1] + log_w + pad_t=0.015	0.8404	0.8340	–
Stage 5b (v3b)	combine_v3b_256	combine_v3a (EMA)	256 (8×8)	2	16×2	1.5e-5 / 2e-6 / 1e-4	Spatial curriculum $7 \times 7 \rightarrow 8 \times 8$; multi-res ensemble [4]	0.8486	0.8384	0.788
Stage 6 (V4)	**v4_control256 + v4_high288**	combine_v3b_256 (RAW)	**256 (8×8) + 288 (9×9)**	**2 + 2**	16×2	**1e-5 / 1.5e-6 / 5e-5**	**Paired 256px + 288px continuations, fixed AdamW weight_decay, partial-accum fix, 4-member common288 ensemble**	**0.8541**	**0.8420**	**0.794**

4.3 Engineering Fixes in V4

In Stage 6 (**V4**), we corrected three subtle training-loop issues present in earlier stages:

1. **AdamW Weight-Decay Key Fix:** Earlier parameter-group dictionaries passed "wd" instead of PyTorch's "weight_decay", causing AdamW to silently fall back to its default 1e-2 weight decay on bias/norm groups. Fixing "weight_decay" restored true zero-decay hygiene on 1D parameters and 1e-4 on 2D weights.
2. **Exact Partial-Batch Gradient Accumulation:** Handled the final partial batch of each epoch ($158 \text{ batches} \equiv 79 \text{ optimizer steps}$) without dropping or mis-scaling trailing gradients.
3. **Ranking-Consistent Checkpoint Selection:** Selected best.pt using the deployment ranking score (obj*q) on both RAW and EMA weights (max(raw_mAP, ema_mAP)).

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5. Post-Processing, Test-Time Augmentation, and Ensembling

5.1 Calibrated Query Scoring (obj*q)

Because the official RetailAction evaluation metric is class-agnostic (c is the maximum score across predicted classes), ranking queries by classification probability suffers from class-split dilution. We rank every query slot by the product of existence probability [5] and expected metric quality [1]:

$$s_q = \sigma\left(z_q^{\text{obj}}\right) \cdot \sigma\left(z_q^{\text{qual}}\right)$$

5.2 Slot-Aligned 4-Pass Test-Time Augmentation (f4 [4])

Because our $Q = 5$ learned query slots maintain stable ordering across augmentations [4], we average predictions slot-by-slot across 4 complementary passes per model:

- s0: Standard 16-frame uniform linspace sampling (frame_shift = 0).
- s1: +1 frame temporal phase shift (frame_shift = 1), reducing temporal discretization error on short actions.
- flip: Horizontal mirror of both camera views and hand tracks (vids.flip(-1)), un-mirroring predicted x-coordinates ($\hat{x} \leftarrow 1 - \hat{x}$) before slot averaging.
- flip1: Combined +1 frame temporal shift and horizontal mirror.

5.3 4-Member Dual-Resolution & Dual-Trajectory Ensemble (high_plus_control)

We combine two paired checkpoints trained from combine_v3b_256 with seed 44:

- **v4_control256/best.pt** (256×256, grid_hw = 8): both **EMA** (c_ema) and **RAW** (c_raw) weight trajectories.
- **v4_high288/best.pt** (288×288, grid_hw = 9): both **EMA** (h_ema) and **RAW** (h_raw) weight trajectories.

All 4 members are evaluated from a unified common_size = 288 loader (each model internally resizes the batch to its native resolution 256 or 288) with equal weights (1 : 1 : 1 : 1, totaling $4 \times 4 = 16$ passes per clip). For each query slot $q \in \{1 \dots 5\}$, temporal bounds, 2D spatial coordinates, and obj*q scores are averaged arithmetically across all 16 renderings.

5.4 Why No NMS + Single Post-Ensemble Temporal Boundary Padding (pad_t = 0.015)

- **No NMS:** Once `obj_head` is supervised across all 5 queries with matched-only class CE [5], duplicate query collapse disappears; applying greedy NMS [3] degrades validation mAP by ~ 1.0 pp by discarding legitimate secondary actions in multi-action clips. We therefore emit all 5 queries per clip (`12,500` actions across `2,500` clips).
- **Post-Ensemble Temporal Padding (`pad_t = 0.015`):** Sigmoid-activated temporal regression heads slightly under-predict start/end boundaries near action onset and completion. Expanding each ensemble-averaged interval symmetrically once by 1.5% of clip duration:

$$[\hat{s}_q, \hat{e}_q] \leftarrow [\max(0, \hat{s}_q - 0.015), \min(1, \hat{e}_q + 0.015)]$$

consistently improves strict temporal AP ($\tau_t = 0.50$).

6. Quantitative Results & Ablations

Method / Configuration	Full Val mAP (1,277)	Val mAPs	Val mAPt	Local Test mAP (2,501)	Test mAPs	Test mAPt	Public LB
Official Host Baseline	—	—	—	—	—	—	0.3910
V2 Baseline (`run2q`, single pass, `cls` rank)	0.6171	0.7610	0.7330	0.5972	0.7420	0.7190	0.5558
+ DIoU + EMA + NMS + `product` (`idea_07` [1, 3])	0.7034	0.8184	0.7874	0.6877	0.8051	0.7742	—
+ STMixer `obj_head` + Per-Layer Re-Matching (`idea_11` [5])	0.7864	0.8647	0.8680	0.7743	0.8519	0.8610	—
+ Dense Per-Frame Head + `f4` TTA + `EMA+RAW` (`combine_v2` [2, 3, 4])	0.8157	0.8840	0.8850	0.8082	0.8717	0.8827	0.7580
+ Pose Cross-Attn + `iou_head` + `256px` Grid (`combine_v3` [1, 6])	0.8486	0.8965	0.9117	0.8384	0.8869	0.9087	0.7882
V4 `256px` Control Only (`c_ema + c_raw`, `f4`, `common288`)	0.8515	0.8961	0.9154	—	—	—	—
V4 `288px` High-Res Only (`h_ema + h_raw`, `f4`, `common288`)	0.8512	0.8977	0.9156	—	—	—	—
V4 Final (`c_ema + c_raw + h_ema + h_raw`, `f4`, `pad_t=0.015`)	**0.8541**	**0.8989**	**0.9176**	**0.8420**	**0.8878**	**0.9127**	**0.7942**

7. Reproducing the `0.7942` Kaggle Submission

From the root of the submission repository:

```
# 1. Install dependencies (or use the provided Dockerfile)
pip install -r requirements.txt

# 2. Download the trained weights zip from Google Drive and unzip
#    so the checkpoints land in weights/ beside the cloned files:
#    https://drive.google.com/file/d/1RXN2al-A_ycb381IqEHMPovggk4NlwM1/view?usp=sharing
#    unzip RetailAction_weights.zip -d weights/
# 3. Place or symlink the official dataset under data/raw/
#    (containing videos/, annotations/, and sample_submission.csv)

# 4. Run the standalone reproduction script using the trained weights in weights/
python reproduce_submission.py --data-root data/raw --out submission.csv
#    (byte-identical to the uploaded Kaggle artifact: combine_ideas_v4_0.7942.csv)
```

The script automatically verifies that `submission.csv` contains all `2,500` challenge clip IDs in exact `sample_submission.csv` order, `12,500` valid actions (`5` per clip), sorted scores in $[0, 1]$, valid coordinates/intervals in $[0, 1]$, and compares against the reference `0.7942` submission.

8. References

[1] Haoyang Zhang, Ying Wang, Feras Dayoub, and Niko Sünderhauf. "VarifocalNet: An IoU-aware Dense Object Detector." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 8514–8523, 2021.

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[3] Chen-Lin Zhang, Jianxin Wu, and Yin Li. "ActionFormer: Localizing Moments of Actions with Transformers." In *European Conference on Computer Vision (ECCV)*, pp. 492–510, 2022.

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